

BB84

Quantum Key Distribution Protocol

Presenters:

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Agenda

- Current Cryptographic Methods
- Threat?
- Primer on Polarization of Light
- BB84 Protocol Steps
- Convention
- Experimental Setup
- Demo
- Qiskit Example

Encryption

Caesars Shift (Shift 3)

Quantum Computers are cool

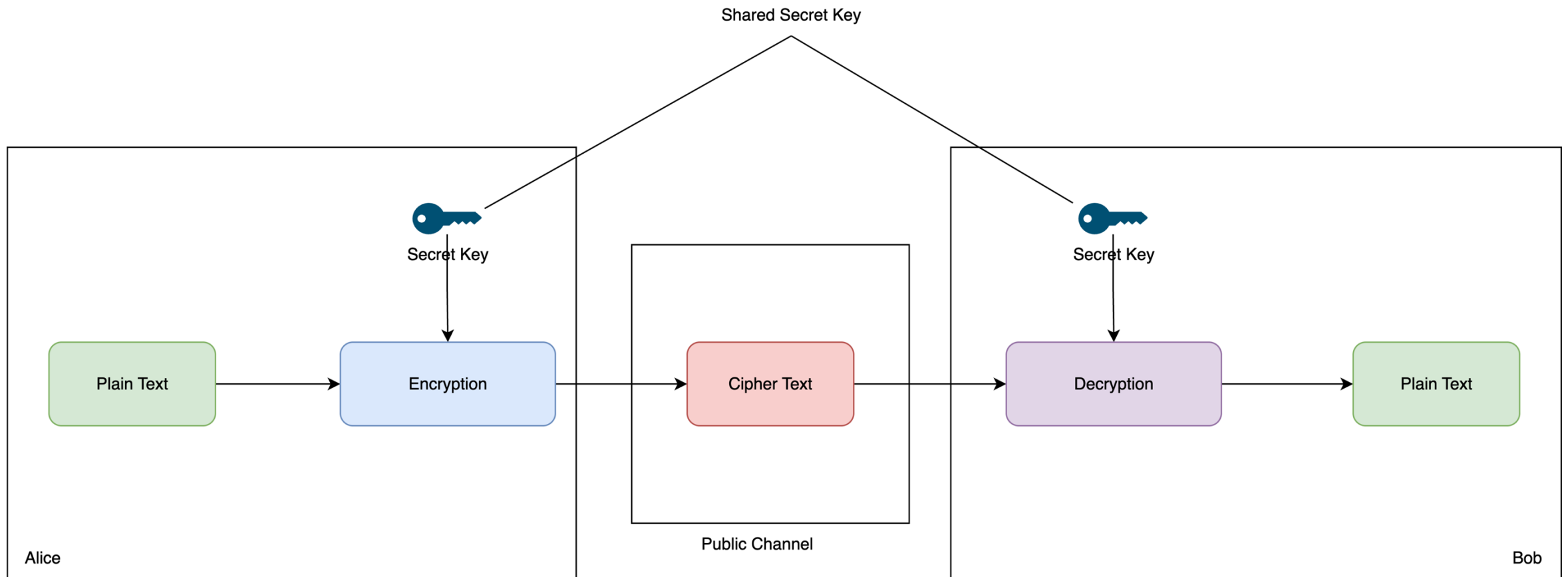
AES

Faa119f6197d2802f50eba03d07a6a766e54c8f39849cc44
7a3a

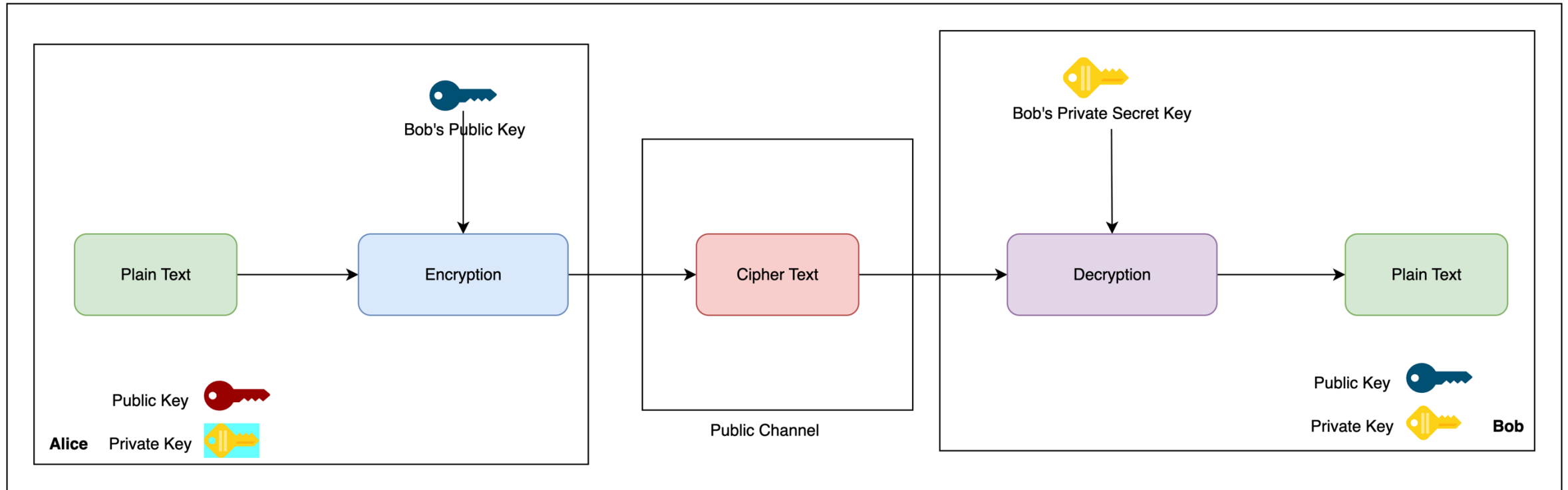
RSA

&\xd0\x02[.sY\xc7\xb0\xfd\xcf\xaa3\x15\xdf\xa9\x99\xce\x
ca\xb9y\xe9x\xf7r%)DoF_\xc7\x04\x94\xbaQ;\xab\xfa\xe8
%\xb6\x94G\xaa\x02\xca\x9f**g6\xd7\x9dC\xf1\xd0w\x1c\
xbc\xc1\x83\x8b\xc7

Symmetric Key Encryption



Asymmetric Key Encryption



Asymmetric Key Encryption

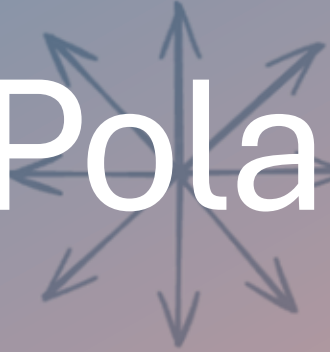
Alice's random bit	0	1	1	0	1	0	0	1
Alice's random sending basis	+	+	×	+	×	×	×	+
Photon polarization Alice sends	↑	→	↘	↑	↘	↗	↗	→
Bob's random measuring basis	+	×	×	×	+	×	+	+
Photon polarization Bob measures	↑	↗	↘	↗	→	↗	→	→
PUBLIC DISCUSSION OF BASIS								
Shared secret key	0		1			0		1

BB84

Quantum Key Distribution

Primer on Polarization

Horizontal Filter



Unpolarized
Light



Horizontally
Polarizing Filter



Vertical Filter

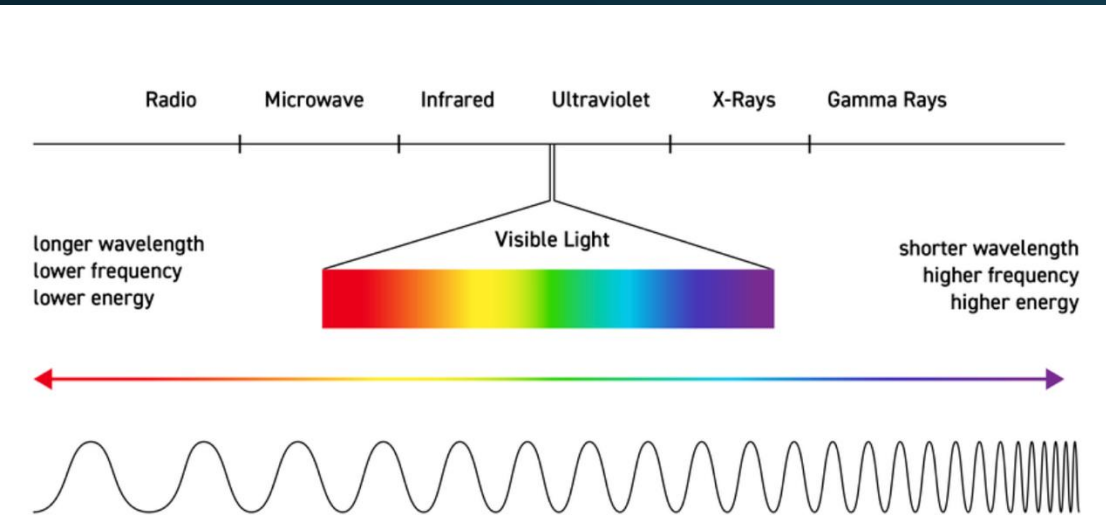


Unpolarized
Light

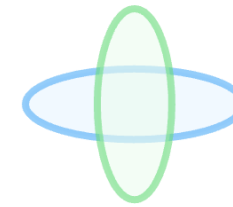


Vertically
Polarizing Filter

Electromagnetic Waves and Polarization



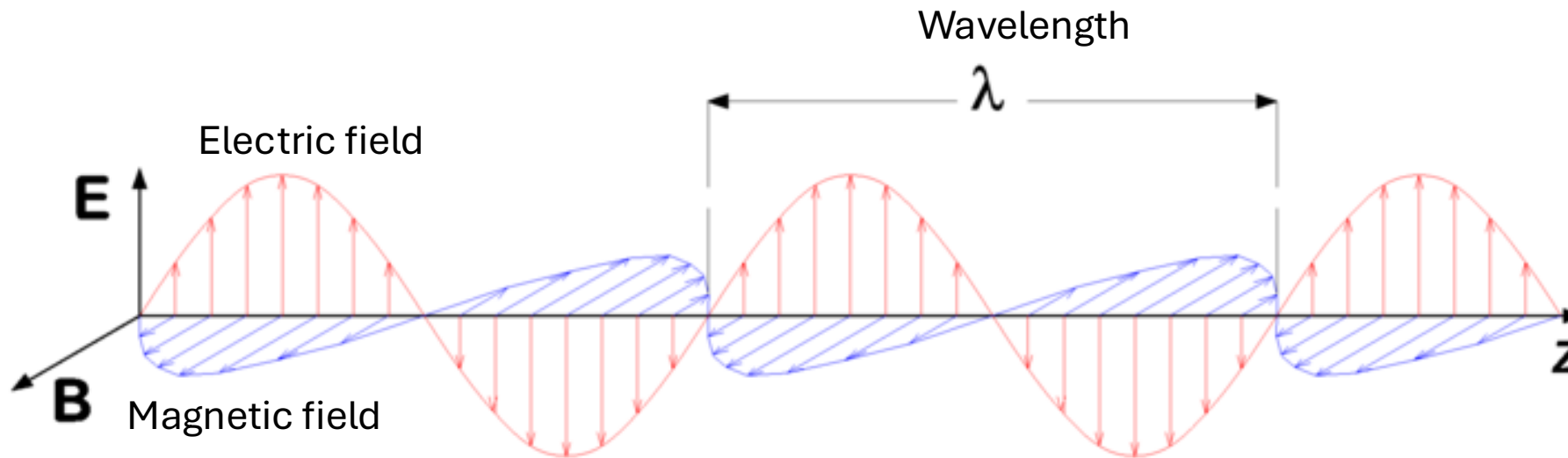
Horizontal



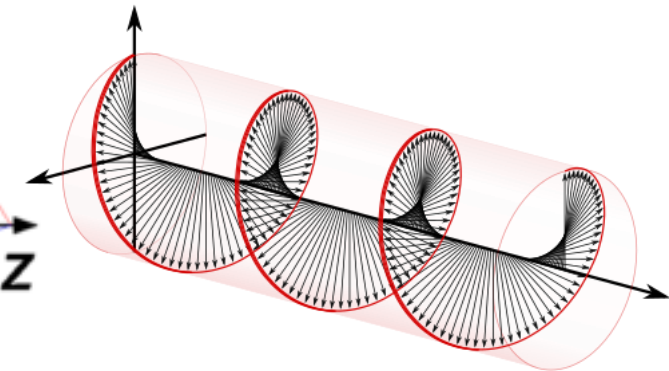
Superposition



Vertical

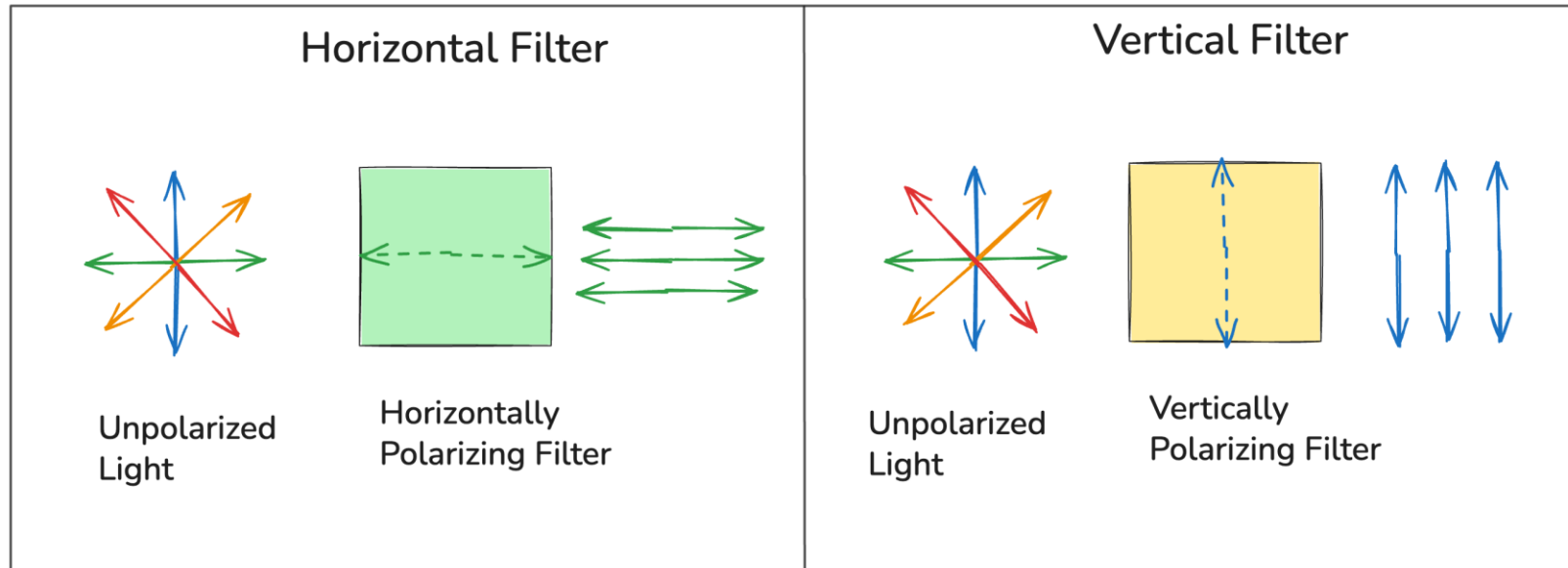


Linear Polarization

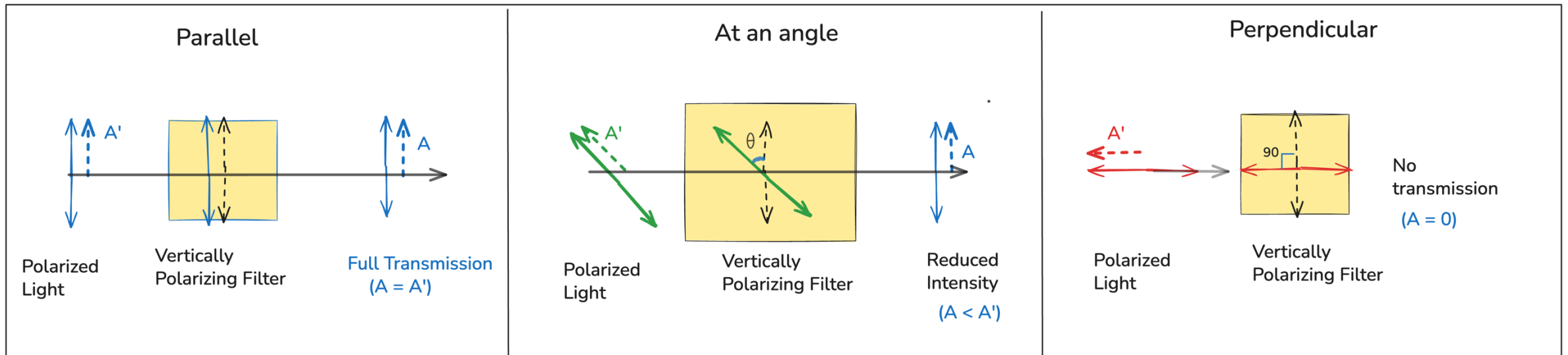


Circular Polarization

Polarizing Filters

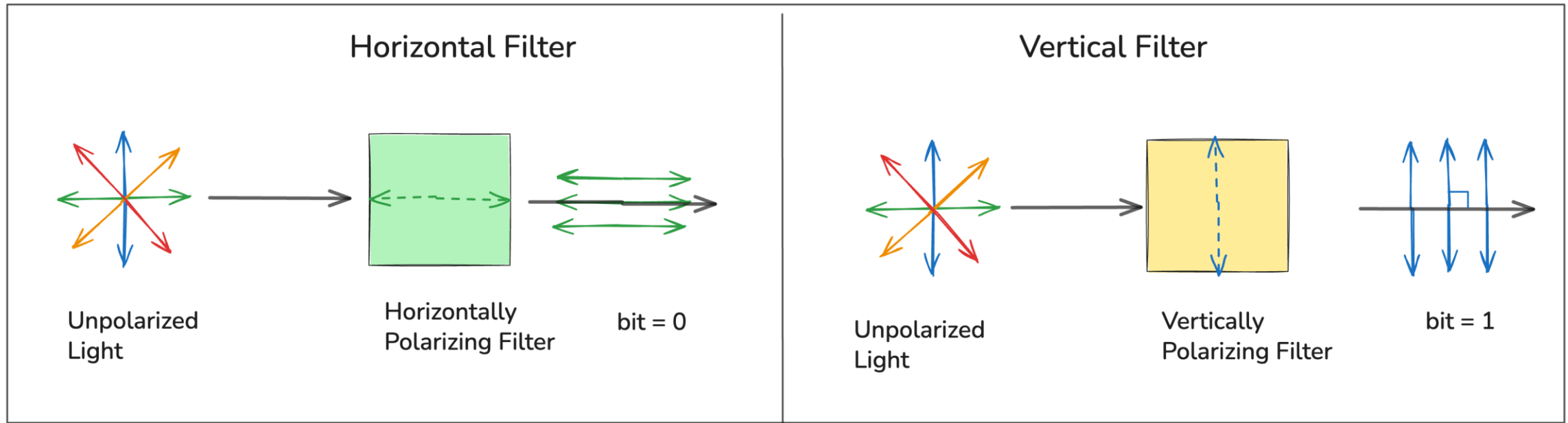


Intensity attenuation



Polarized light incident to linear polarizer at different angles

Convention – Rectilinear (+)

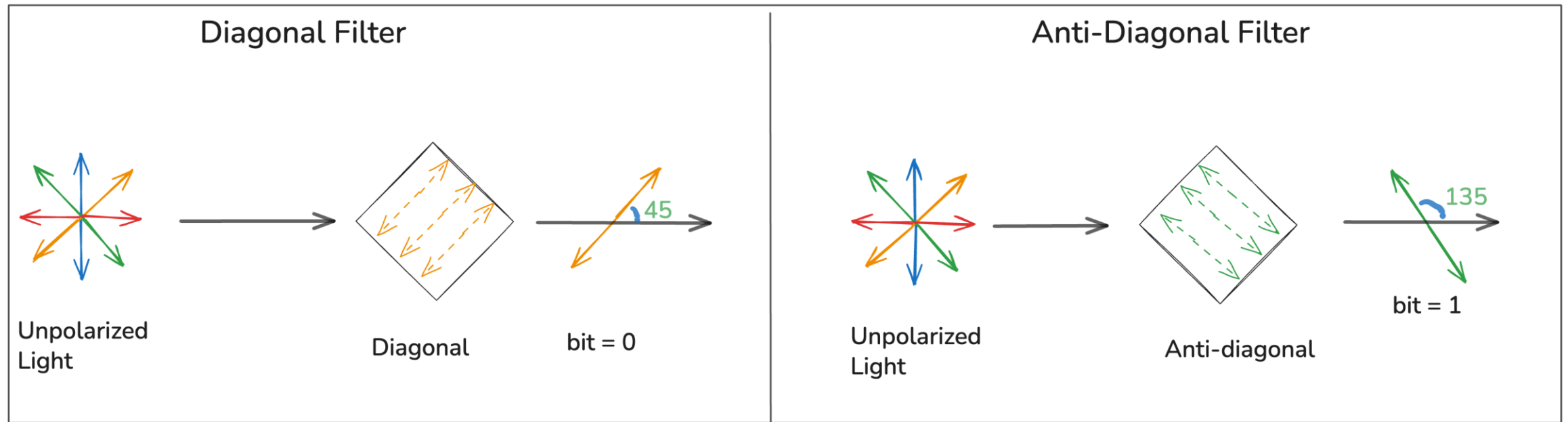


Rectilinear Bases

Horizontal (H) (0°) => bit 1

Vertical (V) (90°) => bit 1

Convention – Diagonal (X)

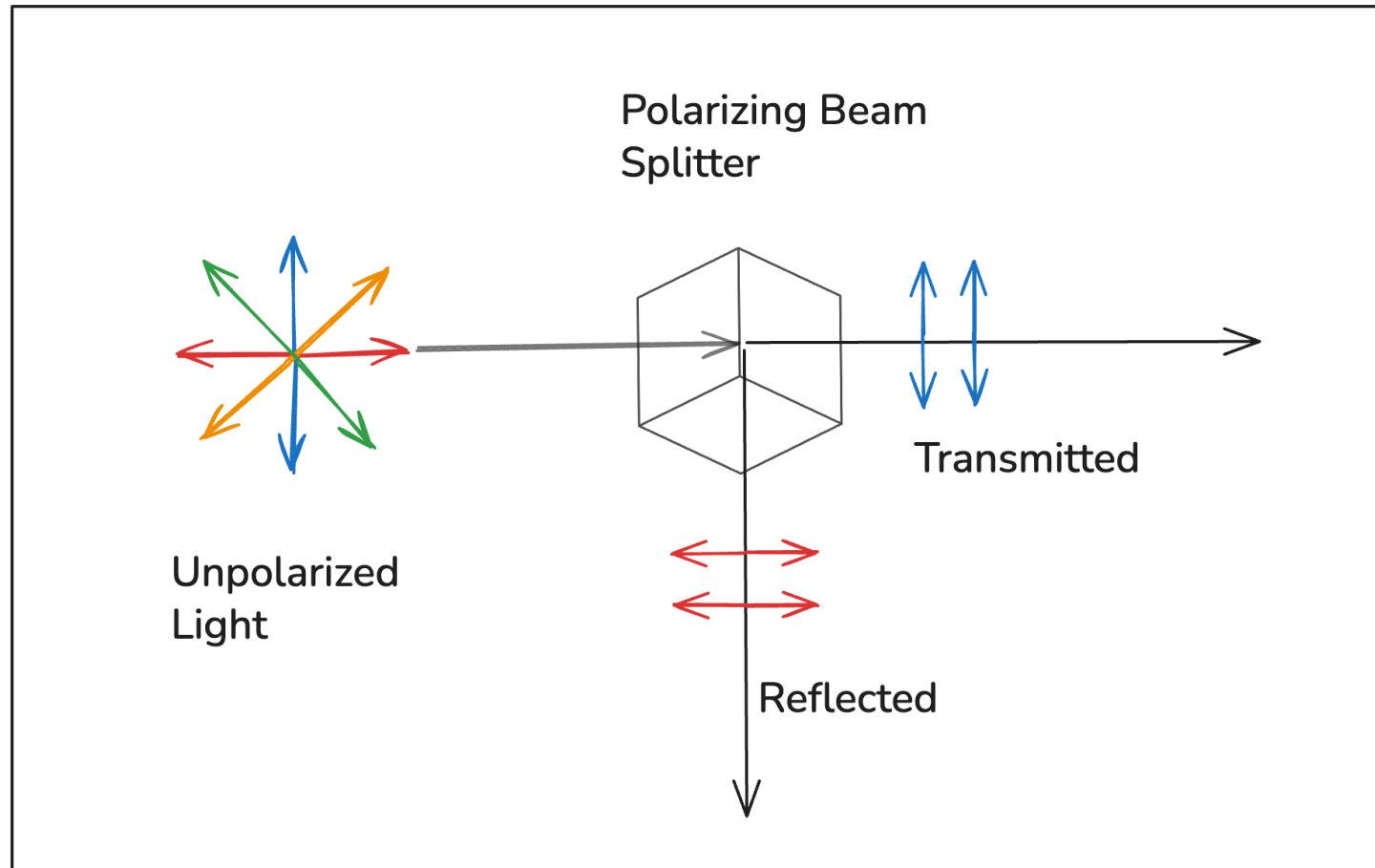


Diagonal Bases

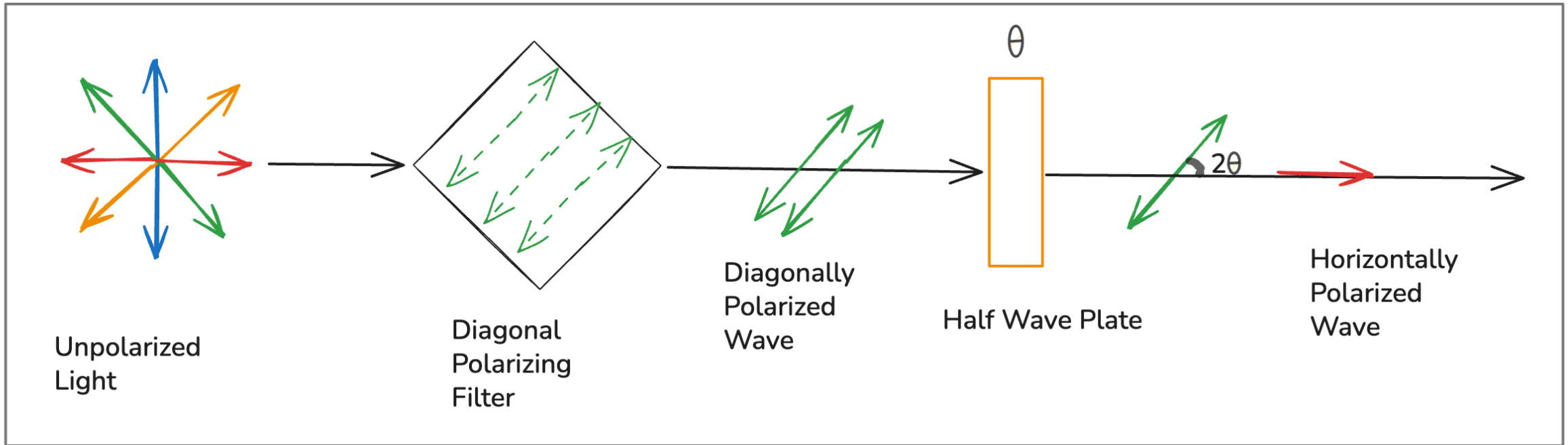
Diagonal (D) (45°) $\Rightarrow$ bit 0

Anti-diagonal (A) (135°) $\Rightarrow$ bit 1

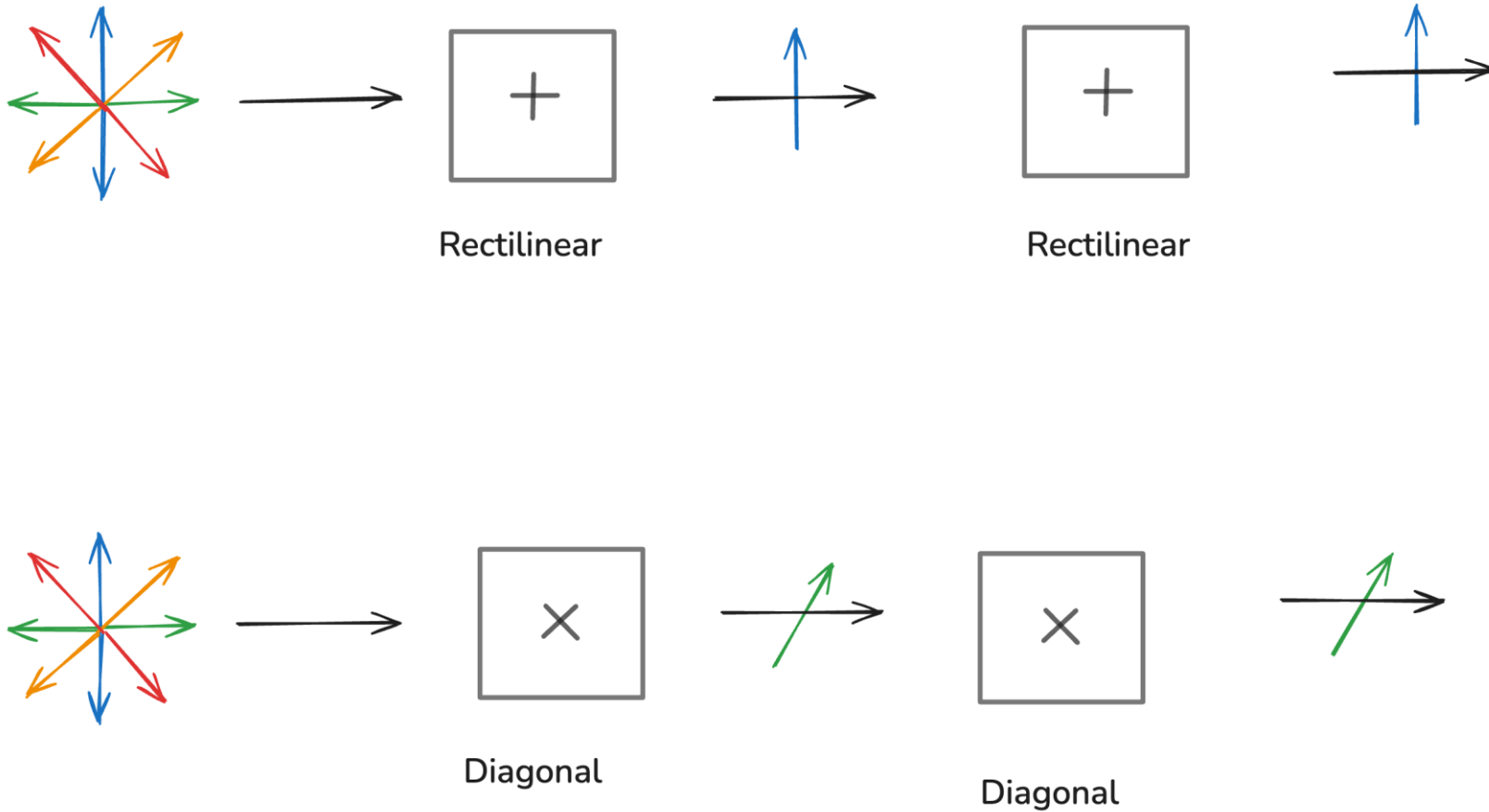
Polarizing Beam Splitter



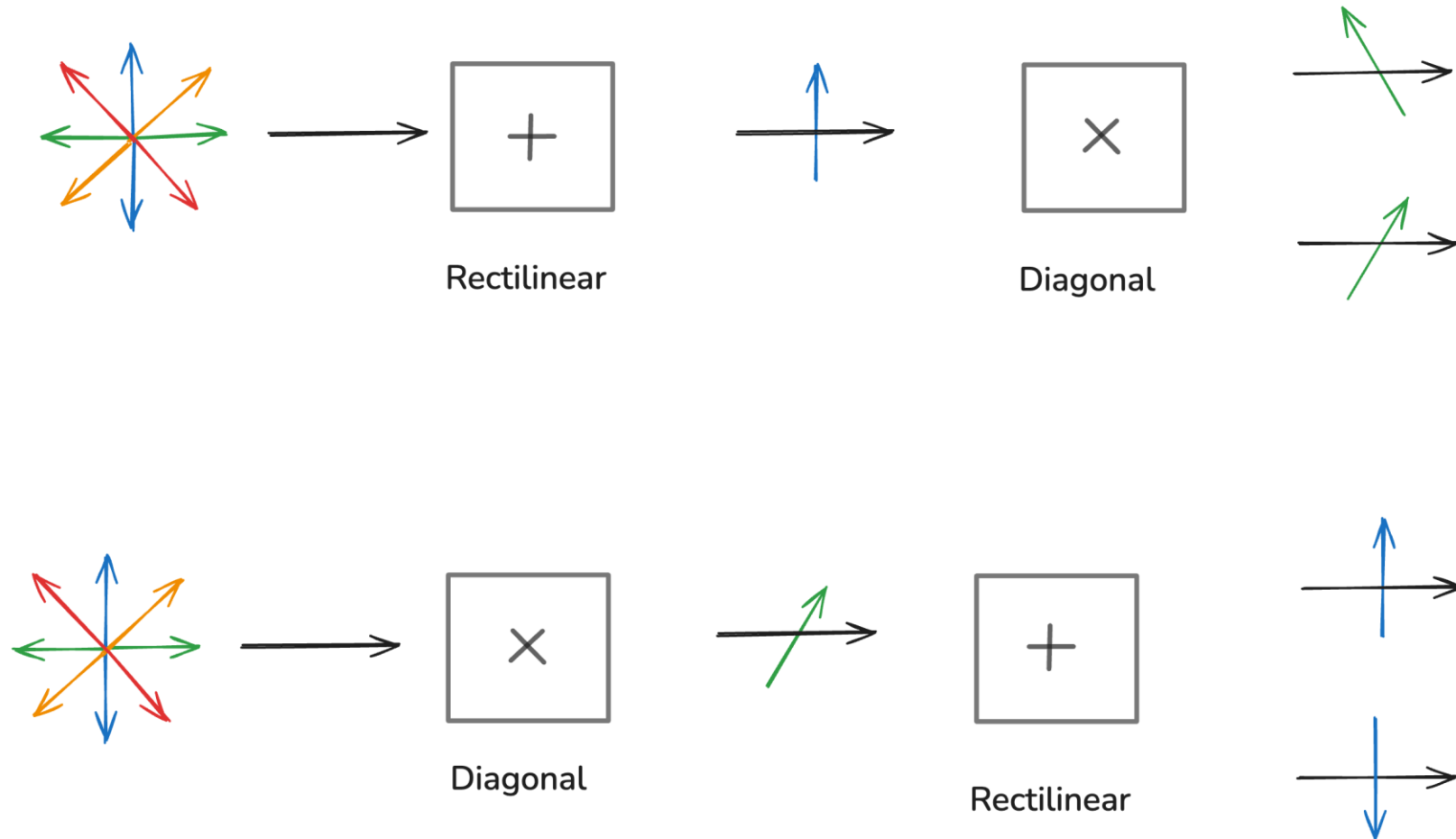
Half Wave Plate



Bases Match

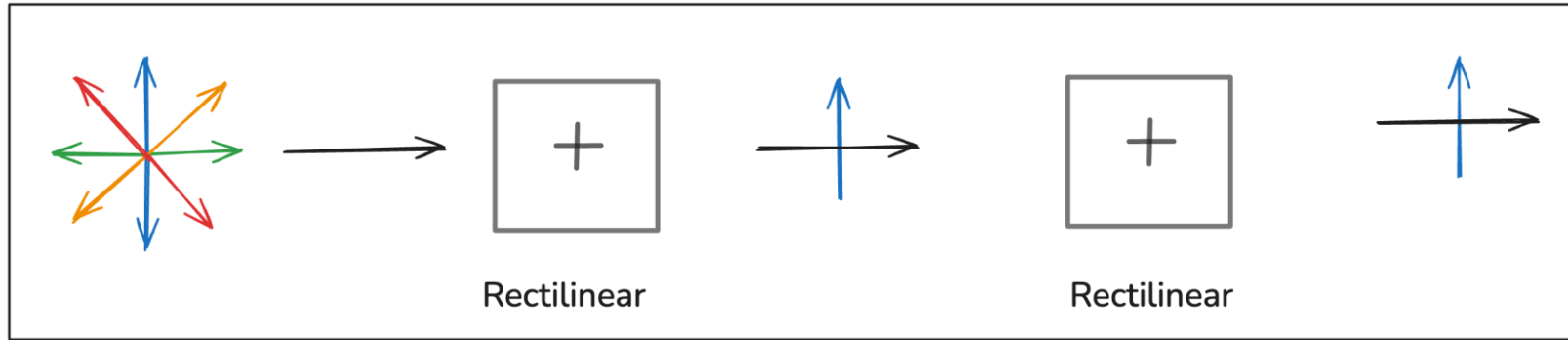


Bases Mismatch



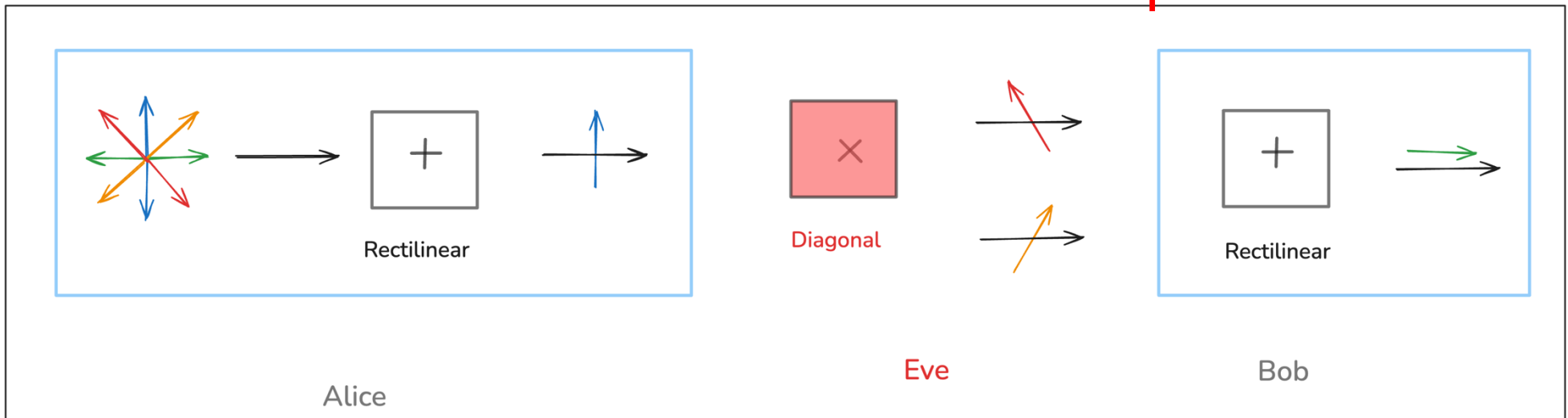
Eve's Interception

No Eve



Cannot intercept undetected

Eavesdropper



BB84

Quantum
Transmission



Key Sifting



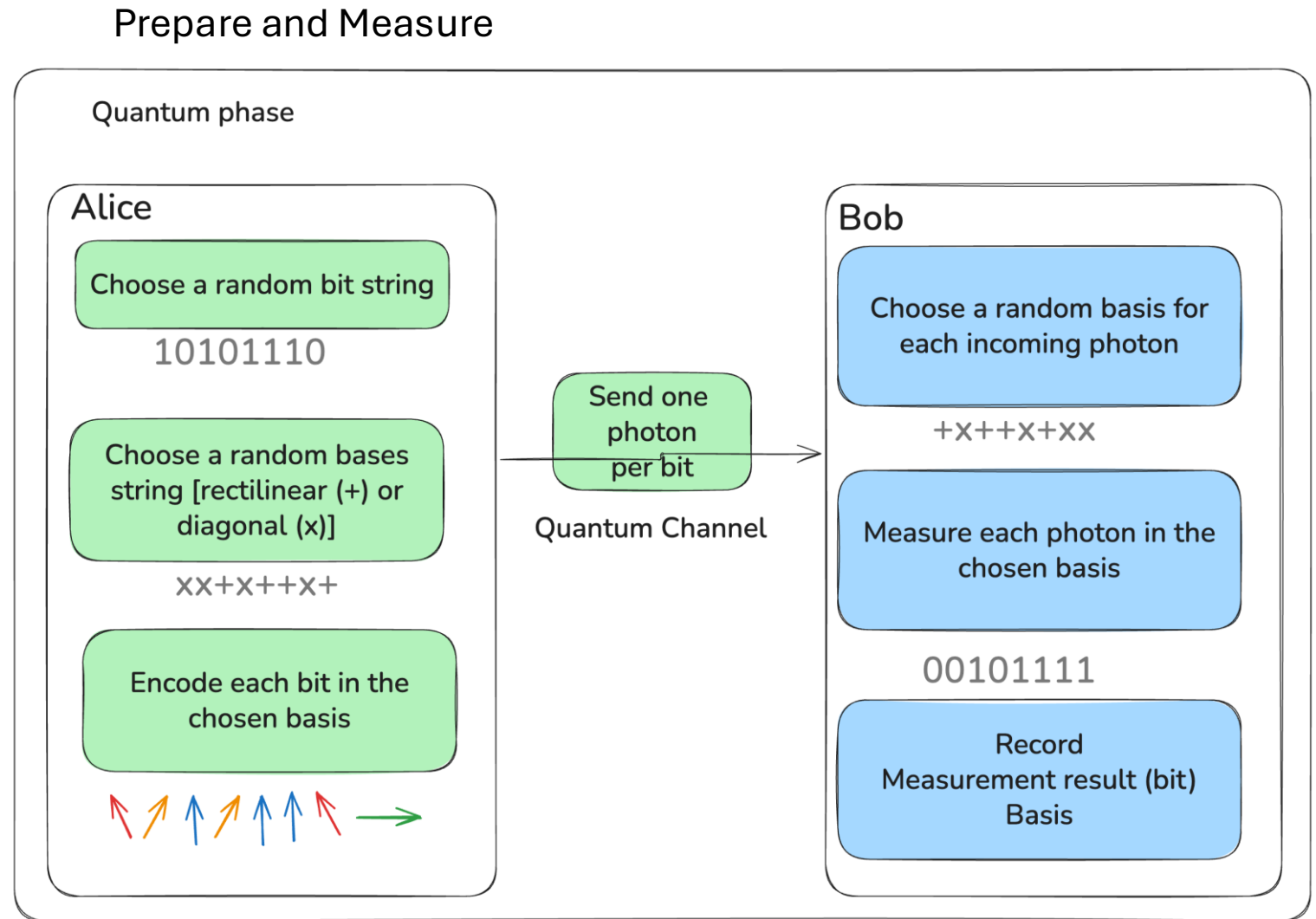
Final Key &
Encryption

BB84

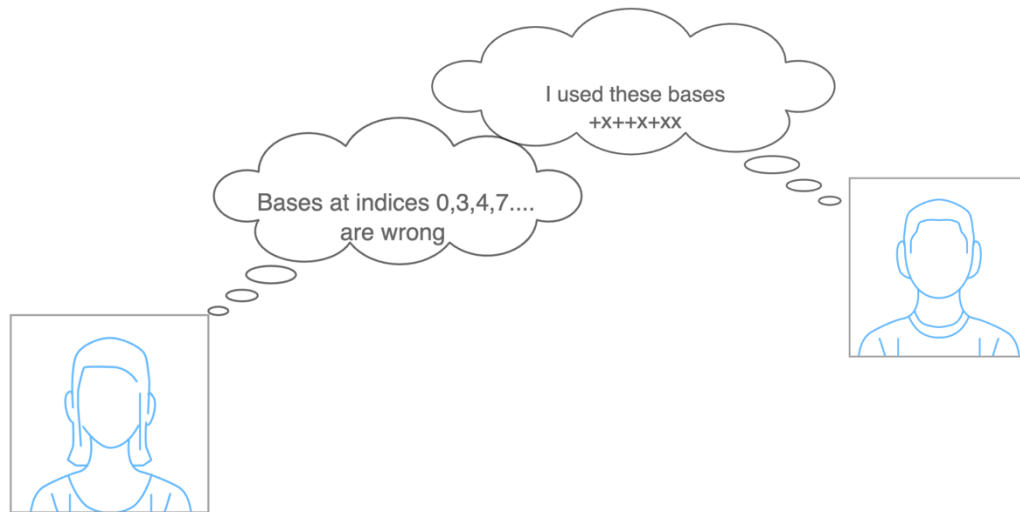
Quantum Key
Distribution

BB84 Protocol

Quantum transmission phase – Prepare and Measure



Classical Post-Processing- Key Sifting



Sifting Phase

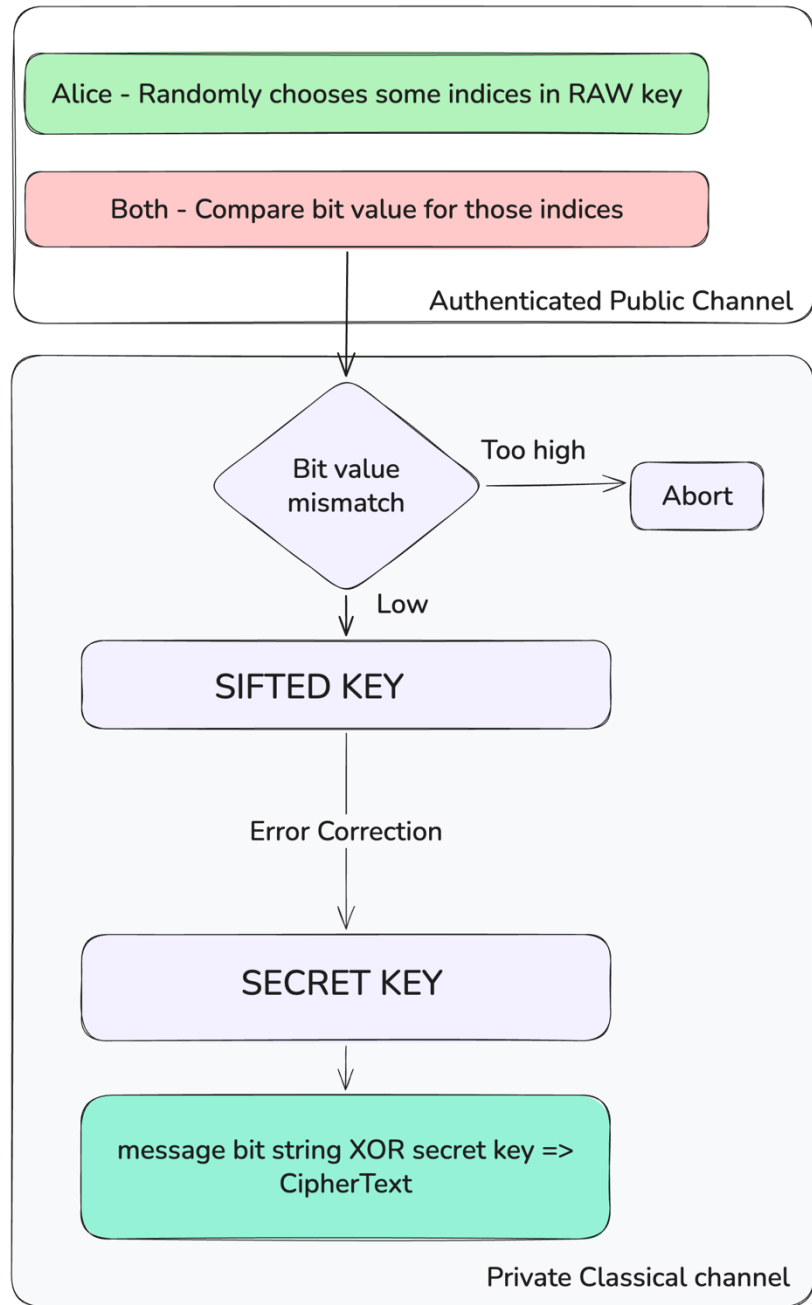
Bob - announce chosen bases (NOT bit value)

Alice - Tell which bases are correct

Both - Discard the mismatched basis positions

RAW KEY

Authenticated Public Channel

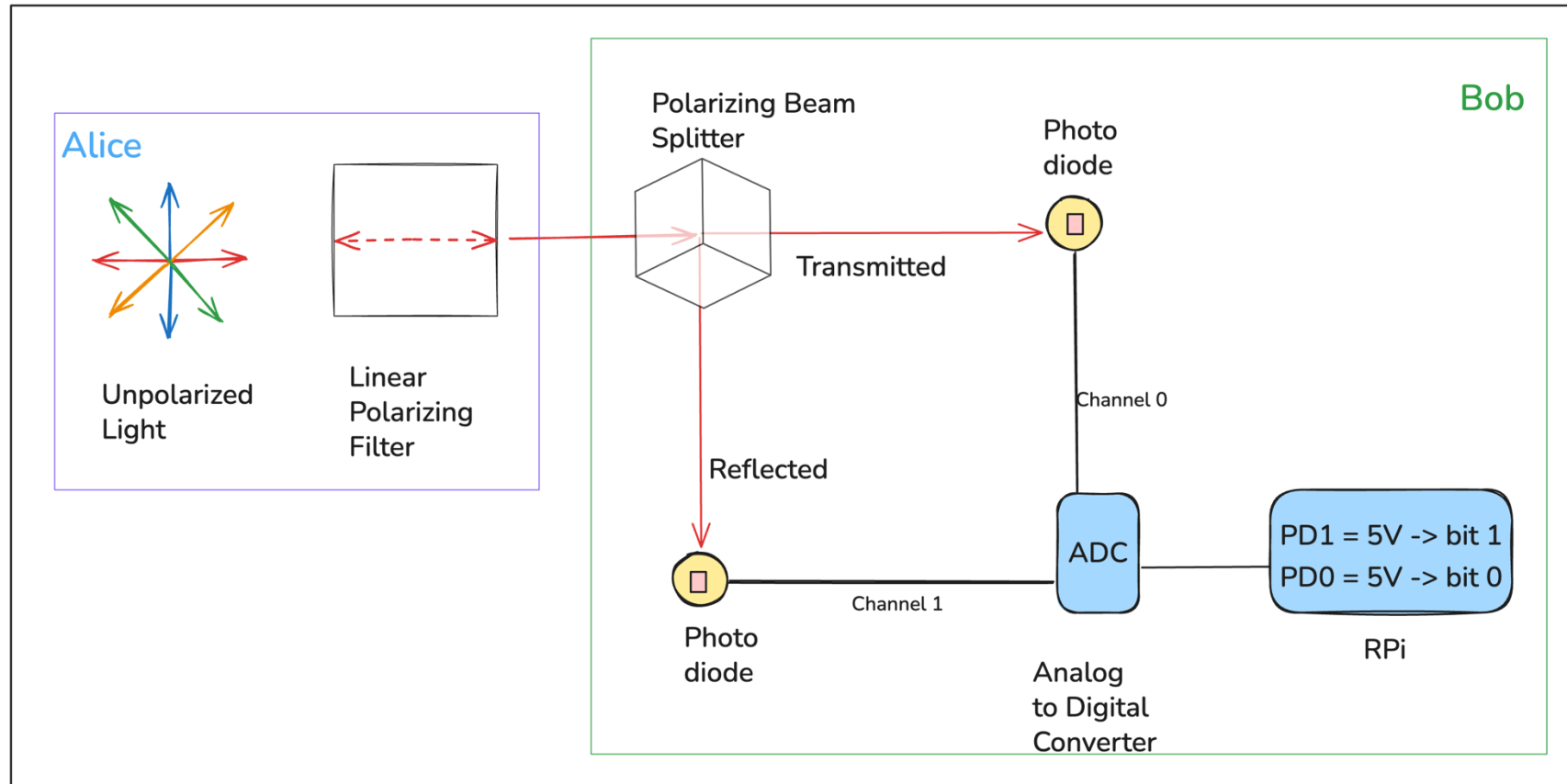


XOR

Message	0	1	0	1
Secret Key	0	1	1	0
Cipher Text	0	0	1	1

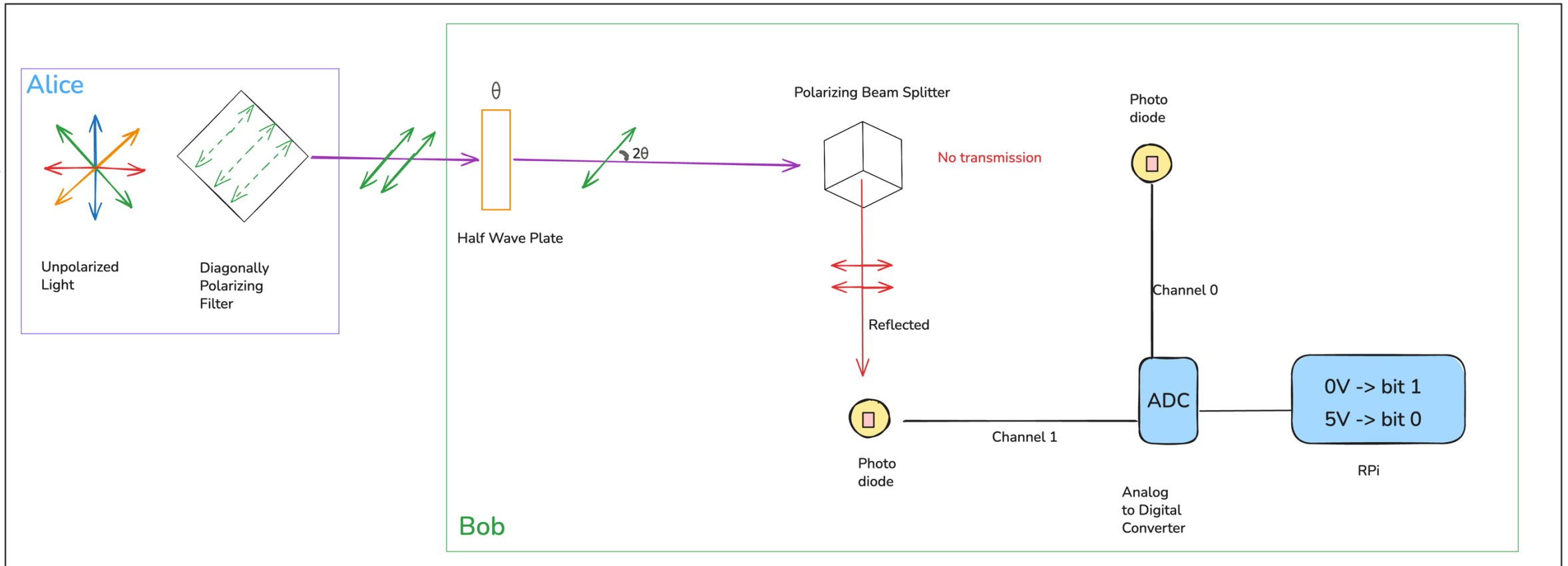
Classical Post-Processing - Final Key & Encryption

Experimental Setup



Experimental Setup

Diagonal Basis Measurement



Diagonal Basis Measurement

Circuit Breadboard with
ADC (Analog to Digital Converter)

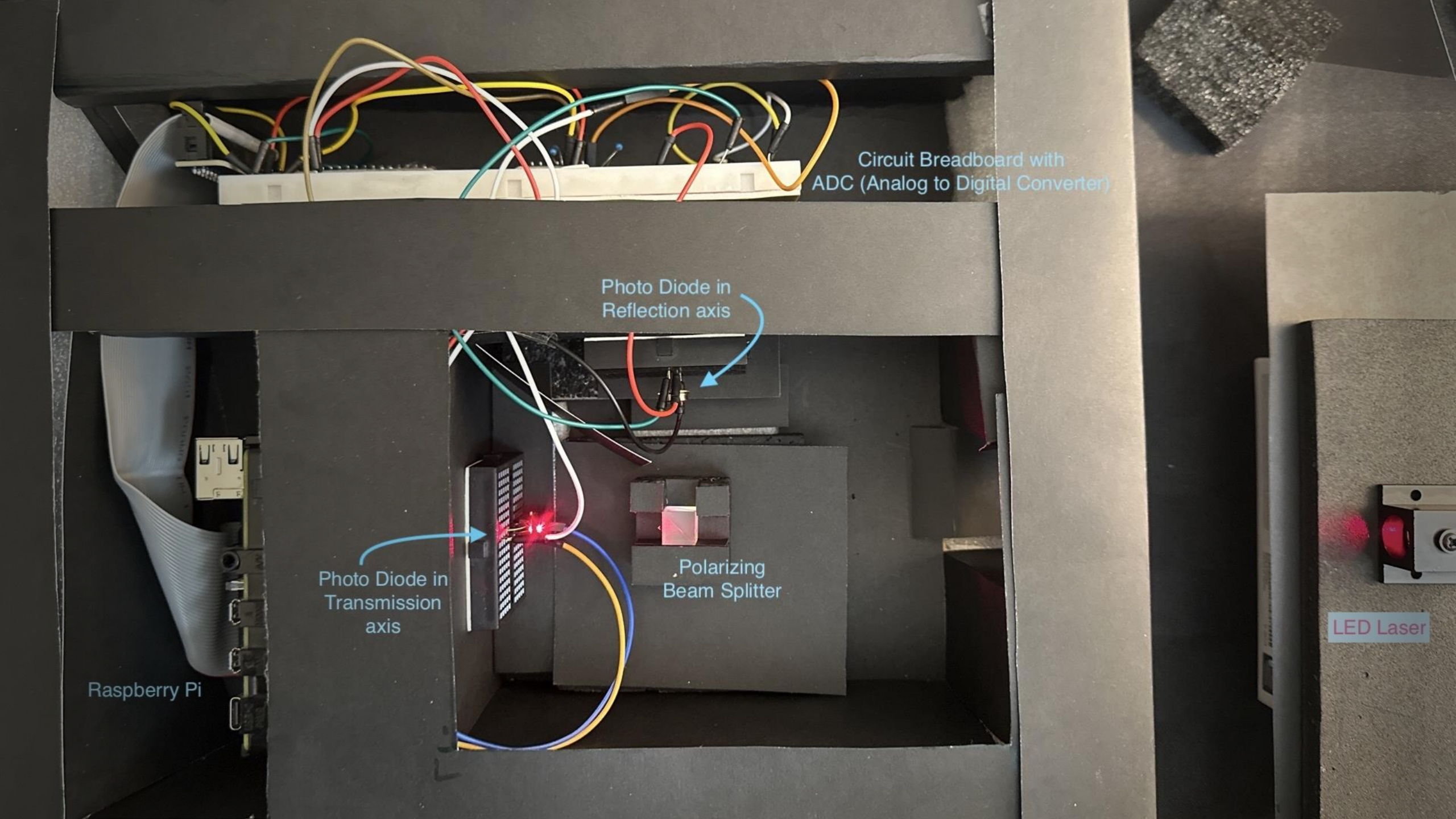
Photo Diode in
Reflection axis

Photo Diode in
Transmission
axis

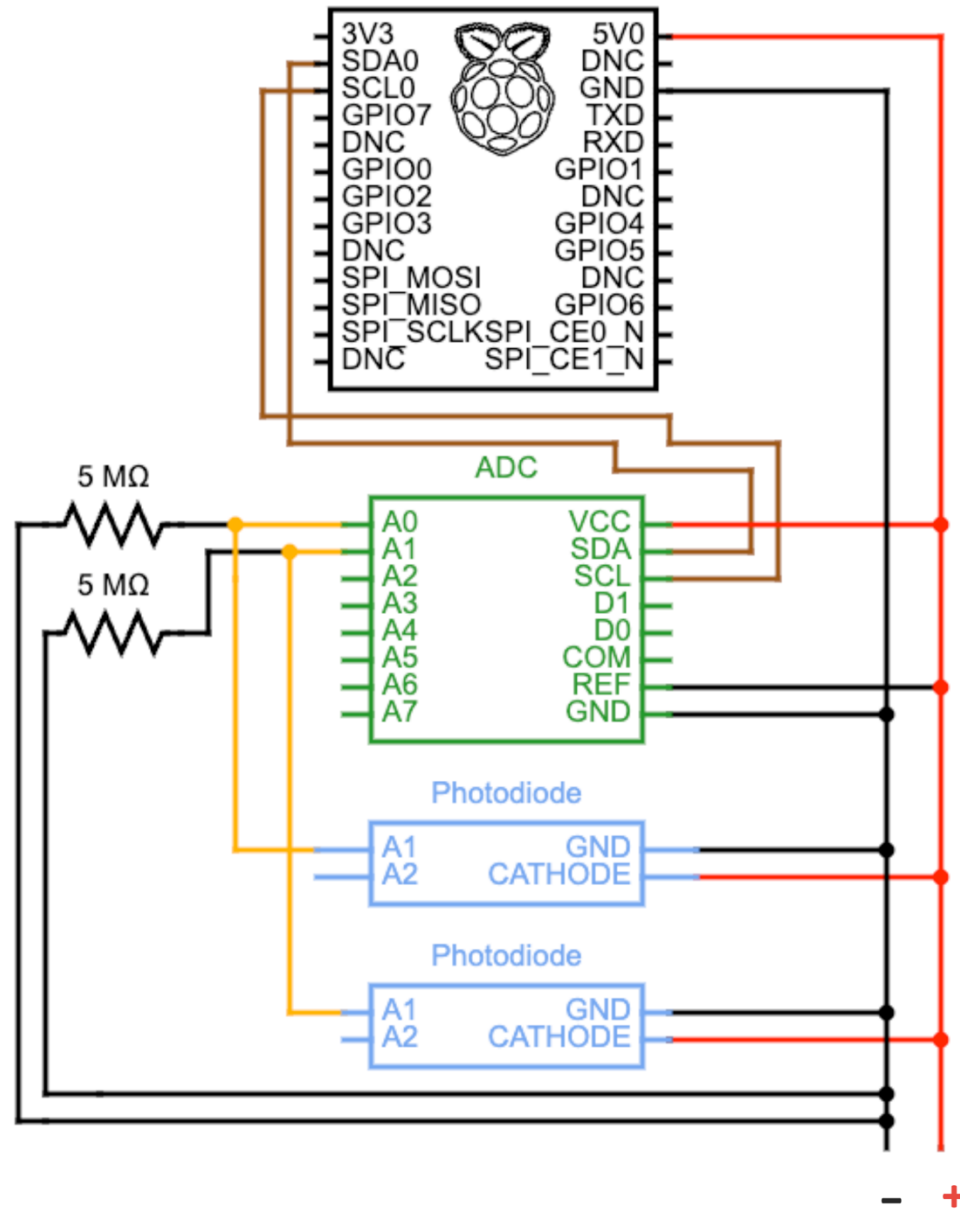
Polarizing
Beam Splitter

Raspberry Pi

LED Laser



Circuit Schematic



BB84 Dashboard – With Eavesdrop

BB84 Quantum Key Distribution

Eve: ☒ ON ☐ Toggle Eve

1 Alice Generate Bits 2 Alice Choose Bases

Alice – Prepared Stream



3 Bob Choose Bases 4 Optical Experiment PhotoDiode 0 voltage: 1 PhotoDiode 1 voltage: 5

Bob – Measurement Stream



5 Key Sifting Compare bases pairs

Sifting



6 Verify bits 7 Final key

Sifted Key



Bit Verification (Public)



Security



Final Secret Key



DEMO

Learning Material

- A practical introduction to quantum-safe cryptography
 - <https://quantum.cloud.ibm.com/learning/en/courses/quantum-safe-cryptography/introduction-to-this-course>
- Quantum key distribution
 - <https://quantum.cloud.ibm.com/learning/en/modules/computer-science/quantum-key-distribution>

Thank You